BigTreeASignalAiBot — Full Feature List

Core Features (Required)

1. RAG Q&A

- Files: bot/rag.py, bot/listener.py, bot/confidence.py
- Description: Users ask questions in Discord channels; the bot retrieves relevant content from the ChromaDB knowledge base and uses GPT to generate answers that imitate the channel owner's style
- Config: OPENAI_API_KEY, LLM_MODEL, RAG_TOP_K, RAG_MAX_DISTANCE, CONFIDENCE_THRESHOLD
- Usage: Automatic — users just post in a target channel

2. Confidence Routing

- Files: bot/confidence.py
- Description: Based on the confidence score, either auto-reply (≥ threshold) or forward to the Owner for review (< threshold)
- Config: CONFIDENCE_THRESHOLD=7 (default), RESPOND_MODE=auto|review|questions|mention_only

3. Owner Review System

- Files: bot/review.py, bot/review_queue.py
- Description: Low-confidence answers are sent to the Owner via DM with Approve / Edit / Reject buttons
- Usage: Owner clicks buttons in DM; Approved content is automatically learned into the knowledge base

4. Knowledge Base Ingestion

- Files: ingestion/ingest.py, ingestion/preprocess.py
- Description: Preprocess, chunk, and vectorize Discord-exported JSON data into ChromaDB
- Usage: python -m ingestion.ingest

5. Channel Listening & Message Handling

- Files: bot/listener.py
- Description: Listen to target channel messages, filter spam/politeness phrases, detect questions, and maintain conversation memory
- Config: TARGET_CHANNEL_IDS, EXCLUDED_CHANNEL_IDS, CONVERSATION_MEMORY_SIZE, CONVERSATION_MEMORY_TTL

Important Features (Strongly Recommended)

6. Rate Limiting

- Files: bot/listener.py
- Description: Per-user cooldown + global per-minute cap to prevent abuse
- Config: USER_COOLDOWN_SECONDS=30, GLOBAL_MAX_PER_MINUTE=10

7. Offline Backfill

- Files: bot/listener.py
- Description: After restart, automatically scan unanswered questions from the offline period
- Config: OFFLINE_BACKFILL_ENABLED=true, OFFLINE_BACKFILL_LOOKBACK_HOURS=24

8. Auto-Learn Owner Messages

- Files: bot/listener.py
- Description: Owner posts in channels are automatically ingested into ChromaDB so the knowledge base keeps growing
- Usage: Automatic — Owner posts normally

9. Thread Reply Support

- Files: bot/listener.py
- Description: Can reply inside Threads and automatically pull Thread context
- Config: THREAD_AUTO_REPLY=true, THREAD_CONTEXT_MESSAGES=15

10. Image / Chart Analysis

- Files: bot/rag.py, bot/listener.py
- Description: When users send images (e.g. candlestick charts), analyze them with GPT-4o Vision
- Config: VISION_MODEL=gpt-4o

11. Voice Message Transcription & Reply

- Files: bot/listener.py
- Description:
- Owner voice: Discord voice clips are transcribed with Whisper and auto-learned into the knowledge base (no public reply)
- Member voice: Only Discord voice clips are processed (not arbitrary mp3/songs). After Whisper transcription, RAG replies; the channel shows typing while transcribing
- Limits: Over ~2MB or 90 seconds prompts the user to use text; after transcription, chit-chat like “thanks/ok” or non-questions are silently skipped without a full reply
- Usage: Send a Discord voice message in a target channel; voice + text caption also works
- Note: Bot must be able to read attachments; transcription depends on OpenAI Whisper

12. Stats Tracking

- Files: bot/stats.py
- Description: Record total queries, auto-replies, forwards, confidence, latency, etc., with time-range filters
- Usage: /stats command or Admin panel

13. Embedding Cache

- Files: bot/cache.py
- Description: LRU + TTL cache for embedding results to reduce duplicate API calls
- Usage: Enabled automatically

Optional Features (Enable as Needed)

14. Admin Web Panel

- Files: bot/admin.py
- Description: Browser-based admin panel for stats / config / knowledge base / FAQ
- Config: ADMIN_ENABLED=true, ADMIN_PORT=8082, ADMIN_SECRET
- Usage: Open http://host:8082/admin/ in a browser

15. FastAPI Management API

- Files: bot/api/ directory
- Description: RESTful API + WebSocket for stats / config / review / knowledge base / FAQ / promo management
- Config: API_ENABLED=true, API_PORT=8090, API_USERNAME, API_PASSWORD
- Usage: Admin mobile app (app/) manages the bot through this API

16. Public Client API

- Files: bot/api/routes_public.py
- Description: Public API for end users (chat, image analysis, FAQ, search, summaries, etc.)
- Config: CLIENT_API_ENABLED=true, CLIENT_API_KEY, CLIENT_RATE_LIMIT_PER_MINUTE=20
- Usage: Web client (web-client/) and mobile client (app-client/) interact via this API

17. Web Client

- Files: web-client/ directory
- Description: React Web App — chat, search, summaries, FAQ, bookmarks, history
- Usage: npm run dev (dev) or npm run build (production)

18. Mobile Client

- Files: app-client/ directory
- Description: Expo React Native App — same features as the Web client
- Usage: Publish to App Store / Google Play via EAS Build

19. Admin Mobile App

- Files: app/ directory
- Description: Expo App for Owner remote management (review, stats, config, knowledge base)
- Usage: Connects through the FastAPI management API

20. Daily Digest

- Files: bot/digest.py
- Description: Daily scheduled 24-hour activity digest posted to a designated channel + DM to Owner
- Config: DIGEST_ENABLED=true, DIGEST_HOUR=22 (UTC), DIGEST_CHANNEL_ID

21. FAQ Auto-Generation

- Files: bot/faq.py
- Description: Cluster high-confidence queries with GPT to auto-generate FAQ entries
- Usage: /generate_faq command or Generate FAQ button in Admin panel

22. Promo Scheduling System

- Files: bot/scheduler.py, bot/commands.py, bot/promo_config.py
- Description: Schedule promo posts / signal recaps / educational content with repeat options (hourly/daily/weekly/monthly)
- Config: PROMO_ENABLED=true, PROMO_CHANNEL_IDS, SIGNAL_PRODUCT_NAME, SIGNAL_PRODUCT_URL
- Usage: /schedule_promo (optional dm_role to also DM the opt-in notify role), /schedule_trial, /schedule_lesson

23. User Testimonial Collection

- Files: bot/testimonials.py
- Description: Auto-detect positive user messages, forward to Owner for Approve/Reject, and display approved testimonials
- Config: TESTIMONIAL_DETECTION_ENABLED=true, TESTIMONIAL_CHANNEL_ID
- Usage: /testimonials to view

24. Webhook Ingestion

- Files: bot/webhook.py
- Description: External systems ingest text into the knowledge base via HTTP POST
- Config: WEBHOOK_ENABLED=true, WEBHOOK_PORT=8081, WEBHOOK_SECRET

25. YouTube Ingestion

- Files: ingestion/ingest_youtube.py
- Description: Ingest YouTube video captions into ChromaDB (captions API first; if none, Whisper fallback: yt-dlp download audio → ffmpeg transcode → OpenAI Whisper)
- Usage: python -m ingestion.ingest_youtube --urls "URL"
- Dependencies: pip install 'yt-dlp[default]' imageio-ffmpeg; if system ffmpeg is missing, place ffmpeg/ffprobe (and deno required by newer yt-dlp) into the venv Scripts/ or bin/ directory. Code resolves binaries by full path and does not rely on system PATH

26. PDF Ingestion

- Files: ingestion/ingest_pdf.py
- Description: Ingest PDF documents into the knowledge base
- Usage: python -m ingestion.ingest_pdf --files "file.pdf"

27. Style Analysis

- Files: ingestion/analyze_style.py
- Description: Analyze Owner writing style and generate a style guide for RAG
- Usage: python -m ingestion.analyze_style

28. Scheduled Ingestion

- Files: bot/ingestion_scheduler.py
- Description: Background scheduled runs of ingestion and style analysis
- Config: INGEST_INTERVAL_HOURS, STYLE_INTERVAL_HOURS

29. Multilingual Support

- Files: bot/config.py
- Description: UI strings in Chinese (zh) and English (en)
- Config: BOT_LANGUAGE=zh

30. Negative Feedback Learning

- Files: bot/review.py, bot/rag.py
- Description: Owner-Rejected answers are saved as negative samples and injected into later prompts to avoid repeating mistakes
- Usage: Automatic; stored in data/negative_samples.json

Slash Commands Summary

Public Commands (All Users)

Command	Description	Usage
/ask	Ask the AI assistant (RAG knowledge base)	/ask question:your question
/signal	View BigTreeSignal product intro (not limited to promo channels)	/signal

/invite	Generate a personal invite link	/invite
/testimonials	Show recent user testimonials	/testimonials
/faq	View FAQ	/faq
/promo_notify	Opt in or out of promo DM notifications	/promo_notify action:opt-in

Owner-Only Commands

Promo management:

Command	Description	Usage
/post_promo	Post a promo immediately (optional sync DM)	/post_promo title:title description:content dm_role:@promo-notify
/schedule_promo	Schedule a promo (supports repeat; optional sync DM)	/schedule_promo title:title description:desc time:time repeat:mode dm_role:@promo-notify
/dm_role	Send promo DMs to the opt-in notify role	/dm_role role:@promo-notify title:title description:content
/promo_notify_panel	Post a promo DM subscription panel in a channel	/promo_notify_panel
/list_promos	List all scheduled promos	/list_promos
/cancel_promo	Cancel a scheduled promo	/cancel_promo promo_id:ID
/schedule_trial	Schedule a free signal recap post (supports repeat)	/schedule_trial title:title content:content time:time repeat:mode

Education management:

Command	Description	Usage
/schedule_lesson	Schedule an educational push (supports repeat)	/schedule_lesson title:title content:content time:time repeat:mode
/list_lessons	List all scheduled lessons	/list_lessons
/cancel_lesson	Cancel a scheduled lesson	/cancel_lesson lesson_id:ID
/resend_summary	Resend a YouTube video GPT summary (auto-ingest if transcript missing)	/resend_summary or /resend_summary video_url:link

Dev / Ops:

Command	Description	Usage
/status	Bot status (uptime, queue, knowledge base count)	/status
/stats	Query stats (totals, confidence, popular questions)	/stats
/search_kb	Search knowledge base documents	/search_kb query:keywords top_k:count
/generate_faq	Auto-generate FAQ from frequent questions	/generate_faq
/weekly_summary	Immediately generate and post this week's summary	/weekly_summary
/daily_summary	Immediately generate and post today's summary	/daily_summary
/views	[Owner] Scan owner posts across the server, briefly summarize, and post to this channel	/views hours:24
/funnel	View acquisition conversion funnel	/funnel days:7

Repeat Mode Options

Schedule commands (/schedule_promo, /schedule_trial, /schedule_lesson) all support these repeat modes:

Option	Description
No repeat	Send once and stop (default)
Hourly	Automatically resend every hour
Daily	Automatically resend every day
Weekly	Automatically resend every week
Monthly	Automatically resend every 30 days

Newer Features

31. Multilingual Auto-Detect & Reply

- Files: bot/lang_detect.py, bot/listener.py
- Description: Auto-detect user message language (zh/en/ja/ko) and instruct the LLM to reply in the same language
- Config: AUTO_LANG_DETECT=true (on by default)
- Usage: Users ask in any supported language; the bot answers in the same language

32. Knowledge Base Quality Report & /kb_report

- Files: bot/commands.py, bot/feedback.py, bot/leaderboard.py
- Description: Owner can view KB quality report: total docs, 30-day queries, low-confidence ratio, satisfaction, frequent questions
- Command: /kb_report

33. User Satisfaction Feedback ([up]/[down] Reactions)

- Files: bot/feedback.py, bot/listener.py
- Description: When users add thumbs-up/down reactions to bot replies, feedback is recorded; Owner can view stats via /satisfaction
- Config: FEEDBACK_ENABLED=true (on by default)
- Command: /satisfaction days:30

34. Conversation Summary & /pin_summary

- Files: bot/commands.py
- Description: Owner uses /pin_summary to summarize recent channel discussion, generate a summary, and pin it
- Command: /pin_summary count:20

35. Scheduled Reminders /schedule_reminder

- Files: bot/reminders.py, bot/scheduler.py, bot/commands.py
- Description: Owner can schedule custom reminder messages (e.g. market open, data releases) with repeat support
- Command: /schedule_reminder, /list_reminders, /cancel_reminder

36. VIP Role Recognition

- Files: bot/listener.py, bot/config.py
- Description: Configure VIP role IDs; users with those roles are exempt from rate limits
- Config: VIP_ROLE_IDS=roleId1,roleId2

37. Keyword Monitoring & Alerts

- Files: bot/keyword_alert.py, bot/listener.py, bot/commands.py
- Description: Owner configures watch keywords; when a user message contains one, the bot immediately DMs the Owner
- Config: KEYWORD_ALERT_ENABLED=true (on by default)
- Command: /add_alert, /remove_alert, /list_alerts

38. Knowledge Base Versioning & Rollback

- Files: bot/kb_versioning.py, bot/commands.py
- Description: KB snapshot management; snapshots can be created before ingest; Owner can browse historical snapshots
- Command: /kb_snapshots

39. Enhanced Welcome Flow

- Files: bot/welcome_flow.py, bot/listener.py
- Description: Multi-step welcome DM for new members: immediate welcome (with /faq /ask /signal). If PROMO_NOTIFY_ROLE_IDS is configured, the same welcome DM includes “Opt in / Unsubscribe” buttons; members only join the promo DM list after clicking
- Config: WELCOME_FLOW_ENABLED=true (recommended), WELCOME_STEP2_DELAY replaced by drip delays: WELCOME_VALUE_DELAY_SECONDS=14400 (value content after 4 hours), WELCOME_CTA_DELAY_SECONDS=86400 (product CTA next day), WELCOME_REMINDER_DELAY_SECONDS=259200 (day-3 reminder)

40. Leaderboard Command

- Files: bot/leaderboard.py, bot/commands.py
- Description: View most active channels, confidence distribution, and other ranked stats
- Command: /leaderboard days:30

41. A/B Test Reply Styles

- Files: bot/ab_test.py, bot/commands.py
- Description: When enabled, each reply randomly picks a style variant and tracks satisfaction per variant
- Config: AB_TEST_ENABLED=false (off by default), AB_TEST_VARIANTS=casual:style_a.txt,formal:style_b.txt
- Command: /ab_results

42. Export Conversations

- Files: bot/export.py, bot/commands.py
- Description: Owner can export conversation logs as JSON or CSV
- Command: /export_conversations format:JSON days:30

Newer Slash Commands Quick Reference

Command	Description	Usage
/schedule_reminder	Schedule a reminder message	/schedule_reminder title:title message:content time:2025-01-01 09:00
/list_reminders	List all reminders	/list_reminders
/cancel_reminder	Cancel a reminder	/cancel_reminder reminder_id:ID
/add_alert	Add keyword alert	/add_alert keyword:keyword
/remove_alert	Remove keyword alert	/remove_alert keyword:keyword
/list_alerts	List monitored keywords	/list_alerts
/kb_report	Knowledge base quality report	/kb_report
/kb_snapshots	View knowledge base snapshots	/kb_snapshots

/leaderboard	Activity leaderboard	/leaderboard days:30
/ab_results	A/B test results	/ab_results
/export_conversations	Export conversations	/export_conversations format:JSON days:30
/pin_summary	Channel discussion summary	/pin_summary count:20
/satisfaction	Satisfaction stats	/satisfaction days:30
/weekly_summary	Immediately generate and post this week's summary	/weekly_summary
/daily_summary	Immediately generate and post today's summary	/daily_summary
/views	[Owner] Scan owner posts across the server, briefly summarize, and post here	/views hours:24
/funnel	Acquisition conversion funnel	/funnel days:7
/invite	Generate a personal invite link	/invite
/promo_notify	Opt in or out of promo DM notifications	/promo_notify action:opt-in
/promo_notify_panel	[Owner] Post promo DM subscription panel	/promo_notify_panel
/dm_role	[Owner] Send promo DMs to opt-in notify role	/dm_role role:@promo-notify title:title description:content
/resend_summary	[Owner] Resend YouTube video summary (auto lookup/ingest)	/resend_summary or video_url:https://youtu.be/xxx

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Environment Variables Quick Reference

All settings are configured via the .env file; see .env.example for a full sample.

Advanced Enhancements

43. Clarification Follow-up After Auto-Reply

- Files: bot/clarification.py, bot/listener.py, bot/config.py
- Description: When confidence is lower but still eligible for auto-reply, ask 1 clarifying question first (e.g. timeframe / risk preference) instead of outputting a vague conclusion
- Config: FEATURE_CLARIFICATION_FOLLOWUP, CLARIFICATION_CONFIDENCE_MAX, FEATURE_CLARIFICATION_CANARY_CHANNEL_IDS

44. Session Memory Summary

- Files: bot/session_summary.py, bot/listener.py, bot/config.py
- Description: In long conversations, automatically compress older messages into a short summary while keeping the most recent turns verbatim, improving context stability and reducing token use
- Config: FEATURE_SESSION_SUMMARY, SESSION_SUMMARY_TRIGGER_MESSAGES, SESSION_SUMMARY_KEEP_RECENT, FEATURE_SESSION_SUMMARY_CANARY_CHANNEL_IDS

45. Feature Flags + Canary Channels

- Files: bot/feature_flags.py, bot/config.py
- Description: Per-channel canary rollout. If the canary list is empty, the feature is enabled everywhere; otherwise only in canary channels
- Example: FEATURE_LANG_DETECT=true, FEATURE_LANG_DETECT_CANARY_CHANNEL_IDS=123,456

46. High-Risk Answer Safety Guardrails

- Files: bot/guardrails.py, bot/listener.py, bot/config.py
- Description: Detect high-risk phrasing (e.g. all-in, guaranteed returns, max position / full portfolio) and either force owner review or append a risk disclaimer template
- Config: FEATURE_SAFETY_GUARDRAILS, GUARDRAIL_MODE=force_review|disclaimer, GUARDRAIL_DISCLAIMER, FEATURE_SAFETY_GUARDRAILS_CANARY_CHANNEL_IDS

47. Active Learning Loop (Feedback-to-Ingestion Pipeline)

- Files: bot/feedback_learning.py, bot/listener.py, bot/review.py, bot/scheduler.py
- Description: Automatically collect [down] feedback and owner-edited replies into a “KB gaps queue,” and daily summarize Top 10 missing questions
- Config: FEATURE_FEEDBACK_LEARNING, FEATURE_FEEDBACK_LEARNING_CANARY_CHANNEL_IDS

48. SLA / Reliability Monitoring

- Files: bot/reliability.py, bot/rag.py, bot/scheduler.py, bot/config.py
- Description: Monitor p95 latency, OpenAI error rate, review queue backlog, scheduler tick lag; alert the owner when thresholds are exceeded
- Config: FEATURE_SLA_MONITORING, SLA_P95_LATENCY_MS_THRESHOLD, SLA_OPENAI_ERROR_RATE_THRESHOLD, SLA_REVIEW_QUEUE_BACKLOG_THRESHOLD, SLA_SCHEDULER_MISS_SECONDS, SLA_ALERT_COOLDOWN_SECONDS

49. Jin10 Market Flash News Feed

- Files: bot/news_feed.py, bot/config.py
- Description: Poll Jin10 Flash API every 30 seconds and push important market flashes (e.g. central bank decisions, NFP, breaking events) to designated Discord channels. Important flashes use Embed rich format; normal flashes are plain text. Supports dedupe, ad filtering, multi-channel push. On bot reconnect, backfills important flashes missed while offline (default last 24 hours, max 50). last_id advances only after a successful post to avoid gaps
- Config: NEWS_FEED_ENABLED=true, NEWS_CHANNEL_IDS=channelId1,channelId2, NEWS_POLL_INTERVAL_SECONDS=30, NEWS_IMPORTANT_ONLY=true, NEWS_BACKFILL_HOURS=24 (if offline longer than 24 hours or more than 50 important flashes, older ones are not backfilled)
- Current routing (2026-09-29): news channel = #时事新闻 (1554564605047476336), Jin10 flashes only; pre-market briefing, macro reminders, corporate announcements and stock movers go to #bigtreesignal_ai (each pinned via explicit channel ID, no longer following the news channel)

50. Weekly Summary

- Files: bot/weekly_summary.py, bot/config.py
- Description: On a configurable weekday at a configured ET time, collect all owner messages and owner replies from designated channels for the week, generate a key summary with GPT, post as Embed to designated channels and @everyone. Reply messages automatically pull the original question as context. Supports multi-channel scan and multi-channel publish. Reliability: channel read / network failures are no longer misclassified as “no messages this week”; retries every 30 minutes after failure; still runs within 18 hours if the scheduled time was missed; sleep uses chunked timing to reduce overslept schedules after PC sleep
- Config: WEEKLY_SUMMARY_ENABLED=true, WEEKLY_SUMMARY_CHANNELS=channelId1,channelId2 (scan sources), WEEKLY_SUMMARY_DAY=5 (0=Mon, 5=Sat), WEEKLY_SUMMARY_HOUR=20 (ET), WEEKLY_SUMMARY_MINUTE=5, WEEKLY_SUMMARY_POST_CHANNELS=channelId (publish targets; empty falls back to SUMMARY_CHANNELS)
- Command: /weekly_summary (Owner manual trigger; can immediately resend last week's missed summary)

51. Promo Auto-Scheduling (Promo Monitor)

- Files: bot/promo_monitor.py, bot/scheduler.py, bot/config.py
- Description: When Owner posts promo info in a designated source channel, the bot automatically cancels old promo schedules, creates a new daily-repeat schedule_promo with @everyone, pushes to target channels at a configured

time (default 4 PM ET), valid for 3 months (configurable). Auto-cancels on expiry. Image attachments are automatically embedded
- Config: PROMO_MONITOR_ENABLED=true, PROMO_SOURCE_CHANNEL_ID=channelId, PROMO_PUSH_HOUR=16, PROMO_DURATION_DAYS=90, PROMO_PUSH_CHANNELS=channelId1,channelId2

52. YouTube New Video Auto-Push (YouTube Monitor)

- Files: bot/youtube_monitor.py, bot/config.py
- Description: Daily at a configured time, poll the YouTube channel RSS feed; on new video, cancel old YouTube lesson schedules and create a new daily-repeat schedule_lesson pushed to Discord channels at a configured time (default 4 PM ET). Push includes title and watch link. If auto-ingest is on, captions/Whisper transcript go into the KB and a GPT summary is pushed to YOUTUBE_SUMMARY_CHANNELS. If transcription fails, a “new video notice” Embed is still sent; already-recorded videos need /resend_summary to resend the summary (see item 56)
- Config: YOUTUBE_MONITOR_ENABLED=true, YOUTUBE_CHANNEL_ID=UCxxxxxxxxxx, YOUTUBE_CHECK_HOUR=11, YOUTUBE_CHECK_MINUTE=05, YOUTUBE_LESSON_PUSH_HOUR=16, YOUTUBE_LESSON_PUSH_CHANNELS=channelId1,channelId2, YOUTUBE_AUTO_INGEST=true, YOUTUBE_SUMMARY_CHANNELS=channelId

53. Daily Summary

- Files: bot/daily_summary.py, bot/config.py
- Description: On each workday (Mon–Fri, configurable) at a configured time, collect that day's owner messages and replies, generate today's key summary with GPT, post as Embed to designated channels and @everyone. Fully independent from Weekly Summary. Posts once per day: if the timer fires early (e.g. 15:59) and already sent, it will not send again in the same minute
- Config: DAILY_SUMMARY_ENABLED=true, DAILY_SUMMARY_CHANNELS=channelId1,channelId2 (scan sources), DAILY_SUMMARY_DAYS=0,1,2,3,4 (0=Mon..6=Sun, default Mon–Fri), DAILY_SUMMARY_HOUR=16 (ET), DAILY_SUMMARY_MINUTE=0, DAILY_SUMMARY_POST_CHANNELS=channelId (publish targets; empty falls back to DAILY_SUMMARY_CHANNELS)
- Command: /daily_summary (Owner manual trigger)

54. Auto Moderation

- Files: bot/auto_mod.py, bot/ban_words.py, bot/config.py
- Description: Automatically detect and delete spam, scam, and ads. Supports 7 detection layers:
1. Keyword matching — zh/en spam/scam/ad phrases (free signals, guaranteed profit, etc.)
2. Ban-word list (exact match) — Owner can dynamically add ban words via /add_ban_word; messages containing them are deleted
3. Ban-word list (semantic match) — OpenAI embeddings for semantic similarity; different wording of the same meaning is still blocked (threshold configurable, default 0.82)
4. External platform links — Telegram, WhatsApp, Line, short links, etc.
5. Discord invite links — discord.gg links from non-exempt users
6. Mass @ mentions / link spam — blocked above threshold
7. Duplicate spam — same user repeating the same content in a short window
- Ban-word management commands:
- /add_ban_word <word— add a ban word (auto-computes embedding for semantic match)
- /remove_ban_word <word— remove a ban word
- /list_ban_words — list all ban words
- Exemptions: Owner (OWNER_USER_ID), Bot, and designated roles (AUTO_MOD_EXEMPT_ROLE_IDS) are exempt
- Logging: Deletes are logged as Embeds to a designated log channel (author, channel, reason, content preview), and Owner is DMed (deleted content + sender Discord ID)
- Config: AUTO_MOD_ENABLED=true, AUTO_MOD_LOG_CHANNEL_ID=channelId, AUTO_MOD_EXEMPT_ROLE_IDS=roleId1,roleId2, AUTO_MOD_MAX_MENTIONS=8, AUTO_MOD_MAX_LINKS=5, AUTO_MOD_DUP_WINDOW=60 (seconds), AUTO_MOD_DUP_THRESHOLD=3,

AUTO_MOD_BAN_WORDS_FILE=data/ban_words.json, AUTO_MOD_BAN_WORDS_SIMILARITY=0.82
- Permission required: Bot needs Manage Messages
- Persistence: Ban-word list stored in data/ban_words.json with words and embeddings; loaded automatically on restart

55. Channel Topic Guard

- Files: bot/topic_guard.py, bot/auto_mod.py, bot/config.py
- Description: Enforce topic limits on designated channels — only allow investment-related discussion (trading, signals, stocks, FX, crypto, technical analysis, etc.). Chit-chat, nonsense, and offensive posts are auto-deleted
- Classification: GPT-4o-mini classifies each message as on_topic (allow), off_topic (delete), or offensive (delete); results cached 5 minutes to avoid duplicate API calls
- Pre-filter: Pure emoji or 1–2 character messages are treated as off_topic without calling GPT
- Fault tolerance: On GPT failure, default to allow (on_topic) to avoid false deletes
- Exemptions: Owner, Bot, and designated roles are exempt (inherits Auto Moderation exemptions)
- Config: TOPIC_RESTRICTED_CHANNEL_IDS=channelId1,channelId2 (empty = disabled)
- Relation to Auto Mod: Topic Guard is the final Auto Moderation layer — after site-wide spam/ban-word filters, restricted channels get a topic check

56. Resend YouTube Summary

- Files: bot/youtube_monitor.py, scripts/resend_youtube_summary.py, ingestion/ingest_youtube.py
- Command: /resend_summary
- Description: Owner uses a slash command to resend a YouTube video GPT summary to YOUTUBE_SUMMARY_CHANNELS. Auto-checks ChromaDB: if transcript exists, summarize directly; if not, Whisper ingest then summarize; if still failing, generate a teaser summary from the title (gold Embed + footer note). Progress updates on the same ephemeral message; send failures are not reported as success. Only one resend job at a time. Works even when Monitor is off. scripts/resend_youtube_summary.py is a CLI backup (ingested videos only)
- Config: YOUTUBE_SUMMARY_CHANNELS=channelId, OPENAI_API_KEY, LLM_MODEL, OWNER_USER_ID
- Whisper dependencies (required when no captions): pip install 'yt-dlp[default]' imageio-ffmpeg; on Windows, place ffmpeg.exe, ffprobe.exe, deno.exe in .venv/Scripts/ (newer yt-dlp needs a JS runtime to download YouTube)
- Usage (Slash command):

/resend_summary — resend the most recently detected video
/resend_summary video_url:https://youtu.be/xxxxx — resend a specific video
/resend_summary video_url:... title:custom title — override title

- Usage (script backup):

python scripts/resend_youtube_summary.py
python scripts/resend_youtube_summary.py --video-id nTWo8Wv7Jao
python scripts/resend_youtube_summary.py --video-id nTWo8Wv7Jao --dry-run

- Flow: ChromaDB has transcript → summarize directly; not ingested → auto Whisper ingest → summarize; transcription fails → teaser summary from title → post to summary channels (with acquisition CTA buttons; see item 59)

57. Purchase Intent Auto-Conversion

- Files: bot/acquisition.py, bot/listener.py
- Description: When users ask about subscription, price, trial, VIP, or how to buy, the bot skips normal RAG and immediately replies with a product Embed + CTA buttons (Learn product / Request trial / View FAQ), and DMs the Owner. Optionally assigns an “Inquiring” role for follow-up
- Config: INTENT_CONVERT_ENABLED=true, INTENT_NOTIFY_OWNER=true, INTENT_LEAD_ROLE_ID=roleId (0 = do not

assign role), SIGNAL_PRODUCT_URL, FREE_TRIAL_ENABLED, FREE_TRIAL_URL
- Usage: Automatic. Placeholder links (e.g. your-product-url.com) will not generate buttons — replace with real URLs

58. New-Member Onboarding DM Sequence (Day 0 / 1 / 3)

- Files: bot/welcome_flow.py, bot/acquisition.py, bot/acquisition_cog.py
- Description: Day-0 welcome DM on join (essentials: #信号说明, /faq, /ask, /signal, plus the 15-day trial notice; Opt in / Unsubscribe buttons when PROMO_NOTIFY_ROLE_IDS is set). Day-1 DM: free trial signal (latest value content). Day-3 DM: paid CTA + testimonials. Jobs persist in data/welcome_drip.json across restarts and are cancelled when a member leaves; VIP role holders stop further drip
- Config: WELCOME_FLOW_ENABLED=true, ONBOARD_DAY1_DELAY_SECONDS=86400 (day 1), ONBOARD_DAY3_DELAY_SECONDS=259200 (day 3)
- Usage: With WELCOME_FLOW_ENABLED on, runs for every new member. Delays are in seconds from join time
- Ops guide: INVITE_REWARDS_TRIAL.md

59. YouTube Summary Conversion Buttons

- Files: bot/youtube_monitor.py, bot/acquisition.py, scripts/resend_youtube_summary.py
- Description: Video summary Embeds include bottom buttons “Learn BigTreeSignal / Request trial / View FAQ,” so educational content also acquires leads. Auto-push and /resend_summary (plus the resend script) all include these buttons
- Config: SIGNAL_PRODUCT_URL, FREE_TRIAL_ENABLED, FREE_TRIAL_URL, YOUTUBE_SUMMARY_CHANNELS
- Usage: Sent with summaries after new-video auto-ingest; resend see item 56

60. Invite Tracking, Leaderboard & Rewards

- Files: bot/acquisition.py, bot/acquisition_cog.py, bot/commands.py, bot/timed_roles.py
- Description: Members use /invite to create a personal Discord invite. On join, the bot diffs invite uses counts and records the inviter. Rewards: ≥ INVITE_ETF_MIN_COUNT (default 5) all-time invites → ETF membership role for 30 days (DM notified, re-earnable after a 30-day cooldown); monthly Top INVITE_TOPN_COUNT (default 5) by trailing-30-day invites → full membership role for 30 days, settled on the 1st (idempotent, retries on failure). Paid cohort members are skipped. Requires Manage Server (list invites), Manage Roles, Create Invite; the bot role must rank above the reward roles
- Config: INVITE_TRACKING_ENABLED=true, INVITE_ETF_REWARD_ROLE_ID, INVITE_ETF_MIN_COUNT=5, INVITE_FULL_REWARD_ROLE_ID, INVITE_TOPN_COUNT=5, INVITE_REWARD_DAYS=30
- Commands: /invite, /invite_leaderboard, /my_invites
- Usage: Members run /invite in any server channel and get an ephemeral personal link
- Ops guide: INVITE_REWARDS_TRIAL.md

61. Conversion Funnel Dashboard

- Files: bot/acquisition.py, bot/commands.py
- Description: Track joins, welcome DM success/denied, /signal uses, purchase-intent hits, CTA sends, invite joins. Data in data/funnel.json
- Command: /funnel days:7 (Owner)
- Usage: Owner privately views last N days (1–90) funnel plus all-time totals. /signal is public and not limited to PROMO_CHANNEL_IDS

62. Opt-In Notify Role Promo DMs

- Files: bot/role_dm.py, bot/commands.py, bot/scheduler.py, bot/welcome_flow.py
- Description: Only send promo DMs to members who voluntarily opted into a notify role. Do not mass-DM manually tagged ops/interest roles (Discord treats unsolicited bulk DMs as spam). Roles outside the whitelist are rejected by the command.
- Config: PROMO_NOTIFY_ROLE_IDS (opt-in “promo notify” role IDs, comma-separated),

PROMO_DM_DELAY_SECONDS=1.2, PROMO_DM_MAX_RECIPIENTS=200
- Usage:
1. Create a “Promo Notify” role in Discord (no admin permissions), put its ID in PROMO_NOTIFY_ROLE_IDS; Bot role must be above it and have Manage Roles
2. Owner runs /promo_notify_panel in an announcement channel (pin recommended); members click “Opt in” to join the list
3. On join, welcome DM also includes the same Opt in / Unsubscribe buttons (requires WELCOME_FLOW_ENABLED=true)
4. Send promos: /dm_role for DMs only (confirm recipient count first); or /post_promo / /schedule_promo with optional dm_role to sync DMs
5. Unsubscribe only removes the notify role; existing ops tags are untouched. Sends exceeding PROMO_DM_MAX_RECIPIENTS are rejected
- Commands: /promo_notify_panel, /promo_notify, /dm_role; dm_role param on /post_promo and /schedule_promo
- Activation & subscription playbook: GROWTH_PLAYBOOK.md

63. Owner Views Summary (/views)

- Files: bot/commands.py, bot/views_summary.py, bot/weekly_summary.py, bot/listener.py
- Description: Owner runs /views in a server text channel or thread; the bot scans the whole server for your recent posts, uses GPT to generate a short, clear key-points summary (about ≤1200 characters, at most 3 subheadings), and publicly posts it in the current channel without @everyone. Collection scope:
- All server text channels / forum posts / public threads: your posts, plus replies to members (reply targets anonymized as “member,” no nicknames)
- Private threads / bot-visible group DMs you participated in: primarily your posts; counterparts keep only very short anonymous context — public summary does not name anyone or paste originals
- Discord does not allow bots to read 1:1 DMs between you and members; to include 1:1 conversations, open a private server thread visible to the bot
- Each channel pages newest→oldest until the time window ends, counting by owner message count so busy channels do not crowd out your posts
- Excluded channels (EXCLUDED_CHANNEL_IDS) and promo source channel (PROMO_SOURCE_CHANNEL_ID) are not scanned
- Usage: /views (default last 24 hours); /views hours:48 (1–168 hours). Owner only; server text channels/threads only
- Command: /views hours:24
- Note: Unlike /daily_summary and /pin_summary: summarizes only the channel owner's views and posts immediately in the command channel

64. Timed Role Grants & 15-Day Read-Only Trial

- Files: bot/timed_roles.py, bot/acquisition_cog.py
- Description: Generic expiring-role engine: grants persist in data/timed_roles.json and are revoked by a 5-minute sweep. Powers (a) invite rewards and (b) the new-member trial: joiners get the 新人体验15天 role (auto-created) with read-only access (view_channel allow, send_messages deny) to member channels, auto-revoked after TRIAL_GRANT_DAYS (default 15). Paid cohort roles (name matching ^懒人.*\d) are protected from the expiry sweep; members who leave get their open grants cancelled. Every privileged Discord call degrades to a warning log without Manage Roles
- Config: TRIAL_ROLE_NAME=新人体验15天, TRIAL_MEMBER_CHANNEL_IDS (comma-separated), TRIAL_GRANT_DAYS=15
- Ops guide: INVITE_REWARDS_TRIAL.md

65. Earnings Digest

- Files: bot/earnings.py
- Description: Each Jin10 poll scans all flashes (bypassing the important-only filter) for earnings releases of watched tickers (must contain hard numbers like EPS/revenue). An LLM extracts EPS vs consensus, revenue vs consensus,

beat/miss %, next-quarter/full-year guidance, and posts a structured Chinese digest to the #财报速读 channel. One post per ticker per earnings season (data/earnings_posted.json dedup). Factual summary only, no trading advice
- Config: EARNINGS_ENABLED=true, EARNINGS_CHANNEL_ID, EARNINGS_TICKERS (comma-separated, e.g. NVDA,AAPL,MSFT,AMZN,GOOGL,META,TSLA,AMD,AVGO,MU,COIN,MSTR)

66. Earnings Calendar Reminders

- Files: bot/earnings.py (EarningsCalendarCog)
- Description: Polls the Nasdaq earnings calendar API every 6 hours; posts a preview 1 day before and a reminder on the morning of each watched ticker's earnings (date, fiscal quarter, consensus EPS with analyst count, before/after market) to #财报速读. Dedup in data/earnings_calendar_reminded.json
- Config: Reuses EARNINGS_ENABLED / EARNINGS_CHANNEL_ID / EARNINGS_TICKERS

67. Stock Movers Alerts

- Files: bot/movers.py (MoversCog)
- Description: During US market hours (Mon–Fri 9:30–16:00 ET) polls Yahoo quotes every 15 minutes; alerts when a watched ticker moves vs previous close past thresholds (default ±3% first alert, ±5% escalation), with related Jin10 headlines as possible reasons. One alert per ticker per threshold per day (data/movers_alerted.json, auto-pruned after 7 days). Skips market holidays via quote-timestamp check to avoid stale-price false alerts
- Config: MOVERS_ENABLED=true, MOVERS_CHANNEL_ID (defaults to news channel; explicitly pinned to #bigtreesignal_ai since 2026-09-29 — movers stay on the signal channel), MOVERS_THRESHOLD_PCT=3.0, MOVERS_ESCALATE_PCT=5.0, MOVERS_POLL_MINUTES=15

68. Portfolio Diagnosis Tool (/持仓诊断)

- Files: bot/portfolio.py (PortfolioCog)
- Description: Members run /持仓诊断 with input like NVDA:30,TSLA:20,AAPL:50 (ticker:weight%, 2–10 holdings). The bot fetches 6 months of daily bars (adjusted closes preferred), computes HHI concentration, average pairwise correlation, annualized portfolio volatility, 6-month return and sector breakdown, then an LLM writes a structured Chinese diagnosis (concentration/correlation/volatility/sector tilt, style summary, risk points). Reply is ephemeral (visible only to the requester); no buy/sell advice; always ends with a not-investment-advice disclaimer
- Usage: /持仓诊断 持仓:NVDA:30,TSLA:20,AAPL:50
- Command: /持仓诊断

69. Pre-market Briefing

- Files: bot/premarket.py (PremarketCog)
- Description: Every trading day at 08:30 ET posts a pre-market briefing to #bigtreesignal_ai: index futures (S&P/Nasdaq) + VIX, today's macro events, today's earnings for watched tickers, and pre-market movers among watched tickers (default ±2%). On market holidays posts a short "market closed" note instead (detected via stale ES futures timestamp). Once per day (data/premarket_briefed.json dedup)
- Config: PREMARKET_ENABLED=true, PREMARKET_CHANNEL_ID (defaults to news channel), PREMARKET_THRESHOLD_PCT=2.0

70. Macro Event Calendar Reminders

- Files: bot/macro.py (MacroCog)
- Description: Built-in calendar of FOMC decisions, CPI, NFP and PCE release dates (verified 2026-09-29 against federalreserve.gov / BLS / BEA official calendars; there is no 2026 US government shutdown — the 43-day Oct–Nov 2025 shutdown's effects must not be applied to 2026). Posts a preview 1 day before and a reminder on the morning of each event to #bigtreesignal_ai. Dedup data/macro_reminded.json (auto-pruned after 7 days). Update the EVENTS list when BLS/BEA publish new dates
- Config: MACRO_ENABLED=true, MACRO_CHANNEL_ID (defaults to news channel)

71. Technical Snapshot (/技术面)

- Files: bot/technicals.py (TechnicalsCog)
- Description: Members run /技术面 with a ticker; the bot fetches 1 year of daily bars and computes RSI(14), MACD(12,26,9), price position vs MA20/MA50/MA200, and latest volume vs 20-day average — objective indicator display only (neutral labels like overbought/oversold; golden/death cross is only labelled on the bar where the crossover freshly forms, otherwise shows DIF above/below DEA), no conclusions, no buy/sell advice. Reply is ephemeral (visible only to the requester) and always ends with a not-investment-advice disclaimer
- Command: /技术面 代码:NVDA

72. Quick Quote (/报价)

- Files: bot/technicals.py (TechnicalsCog)
- Description: Members run /报价 with a ticker; returns last price, % change vs previous close, open/high/low/volume, plus pre/post-market price when available. Reply is ephemeral (visible only to the requester) and always ends with a not-investment-advice disclaimer
- Command: /报价 代码:NVDA

73. Corporate Announcement Monitor

- Files: bot/announcements.py (hooked into the NewsFeedCog flash scan chain)
- Description: Scans all Jin10 flashes on every poll (not limited by the important-only filter); when a watched ticker has a buyback, secondary offering/dilution, or analyst rating change (upgrade/downgrade/target-price change), posts a short alert to #bigtreesignal_ai with the flash excerpt. Tickers match as standalone tokens (MU won't hit musk, COIN won't hit bitcoin); central-bank reverse-repo ops don't count as "buybacks". Earnings-type flashes are handled by the earnings digest and skipped here to avoid duplicates. Dedup by flash ID (data/announcements_posted.json)
- Config: ANNOUNCE_ENABLED=true, ANNOUNCE_CHANNEL_ID (defaults to news channel), reuses the EARNINGS_TICKERS watchlist

74. DM Broadcast Outreach 【Feature Highlight】

- Files: bot-ops/dm_broadcast.py (standalone script, does not depend on the live bot), bot/trial_cog.py (button interactions)
- Description: Server announcements (even @everyone) only reach a few dozen members in practice. This is the standard playbook for "getting one promo message to every member": the bot DMs every member individually with a button — one click to join. Four-layer reach: ① batched DMs (primary; throttled 8–12s per message, max 400/day, ~5,000 members in ~10 days, 429 backoff per Retry-After); ② behavior-triggered DMs (on join / first message / first reaction, highest open rate); ③ server announcement as backup (covers members with DMs closed); ④ ambassador word-of-mouth. Compliance guardrails: every DM carries an opt-out (reply STOP to unsubscribe permanently), max 1 broadcast + 1 triggered DM per topic per user. Resumable state file, idempotent; DM-closed members are exported to a separate list for announcement follow-up. See docs/zh/growth/DM_OUTREACH_PLAN.md
- Config: the standalone script reads the token from ~/workspace/user/files/env.txt; --dry-run previews without sending
- Usage: python3 bot-ops/dm_broadcast.py --dry-run to preview → python3 bot-ops/dm_broadcast.py for the real batched send

75. One-Month Free Trial (信号体验 / Trial30)

- Files: bot/trial_cog.py, bot/trial_config.py
- Description: One-month free signal trial campaign. Members run !试用 in the server or click the button in the promo DM → the bot checks eligibility (non-paying member, once per lifetime, Discord account age ≥ 30 days) → immediately grants the "信号体验" role (read-only on 7 member channels, valid through 2026-12-31) → 30-day timer starts. The bot then DMs the user to DM the owner with a share screenshot + TradingView username; the owner manually grants the ETF indicator permission on TradingView and marks it with !tvdone @user <TV_username>. Six conversion drips (Day 1/7/14/21/25/30) nurture toward paid conversion; on day 30 the role is auto-revoked and an "体验过期" (trial-expired) marker is applied. Kill switch TRIAL30_ENABLED (default false)

- Config: TRIAL30_ENABLED=true, TRIAL_CAMPAIGN_END=2026-12-31, TRIAL30_INVITE_LINK
- Commands: !试用 (members), !tvdone @user TV_username (admins)

76. bigTreeSignal Credit Ledger

- Files: bot/credit.py (pure ledger logic), bot/credit_cog.py (commands), bot-ops/import_credits.py (bulk import script)
- Description: Subscriptions are collected manually (PayPal/Alipay/WeChat Pay), so promo credit needs a dispute-proof ledger. Keyed by Discord ID (optionally linked to a TradingView ID); every grant/redemption is journaled (timestamp, amount, reason, operator) with balances to 2 decimals. Admin commands: !grant_credit @member amount reason (issues credit, DMs the user), !use_credit @member amount reason (redeems toward renewal; fails cleanly on insufficient balance), !link_tv @member TV_username (links a TradingView ID); self-service: /my_credit (ephemeral, shows balance + TV link + last 5 entries). After a promo the owner supplies an ID list (CSV: discord_id,tv_username,amount,reason) and import_credits.py --dry-run previews the bulk import
- Config: none (CREDIT_STATE_FILE env override exists for tests only)
- Commands: !grant_credit / !use_credit / !link_tv (admins), /my_credit (all members)